Obtaining the Cosmic Energy Budget from the CMB Observations

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Certificate

This is to certify that the project titled *"Obtaining the Cosmic Energy Budget from the CMB Observations"* has been conducted by Dhriti Kotwal for the VSP project, as stipulated by IIA, Bengaluru. The project was performed under my guidance. I certify that the project is her bona fide work and has been carried out to my satisfaction.

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Place: Bangalore

Declaration

I, Dhriti Kotwal, declare that, to the best of my conscience, the project entitled *"Obtaining the Cosmic Energy Budget from the CMB Observations"* was performed by me under the guidance of Dr. Suvedha Suresh Naik.

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Abstract

The Cosmic Microwave Background (CMB) radiation we observe today is a snapshot of the universe around 380,000 years after the Big Bang. The Planck satellite has measured the variations in the brightness of the CMB across the sky with unprecedented precision. These data contain rich information about the universe. One such key information is the energy budget of the universe. In this project, we have used the CMB data from the Planck archive. In order to understand how the cosmological parameters, affect the CMB observations, we use the code CLASS and simulate the CMB angular power spectrum for different combinations of cosmological parameters. Combining the simulated outputs with the Planck data, we derive the best-fit Lambda-CDM model and hence determine the current energy budget of the universe.

Contents

1 Introduction **9**

2 Theory **11**

2.1 The ΛCDM Model 11

2.2 Timeline of Key Cosmic Events[4] 11

2.3 Brief Overview of the CMB Power Spectrum [5, 6] 12

3 Methodology **15**

3.1 Data 15

3.2 Codes 15

3.3 Statistics 17

4 Results **19**

4.1 Section A: Varying the Parameters 19

4.1.1 Baryonic Density $\Omega_b h^2$ 19

4.1.2 Cold Dark Matter Density $\Omega_{\rm cdm} h^2$ 21

4.1.3 Hubble Constant H_0 22

4.1.4 Reionization Optical Depth τ 23

4.1.5 Linear Matter Fluctuation Amplitude σ_8 24

4.1.6 Scalar Power-Law Index n_s 25

4.2 Section B: Finding Ω_b Using χ^2 Statistics 26

5 Summary and Future work **29**

Chapter 1

Introduction

The Cosmic Microwave Background (CMB) is the afterglow of the Big Bang — the oldest light we can observe, released when the universe was around 380,000 years old. At that time, the universe had cooled enough for protons and electrons to combine into neutral hydrogen, making the universe transparent to radiation. This light has been travelling freely ever since, redshifted by the expansion of the universe to microwave wavelengths. Today, it appears as a nearly uniform blackbody with a temperature of about 2.73 K, filling all of space.

The existence of the CMB is one of the strongest pieces of evidence supporting the Big Bang theory. Its detection in 1965 by Arno Penzias and Robert Wilson, who stumbled upon a uniform microwave noise while working with a radio antenna, confirmed earlier theoretical predictions made in the 1940s by George Gamow and collaborators. Since then, several experiments — including COBE, WMAP, and Planck — have mapped the CMB with increasing precision. These maps show tiny temperature fluctuations, or anisotropies, at the microkelvin level, carrying a wealth of information about the early universe.

These small fluctuations are not random noise — they are imprints of sound waves in the early universe, caused by interactions between photons and baryons. The detailed statistical properties of these anisotropies, especially as captured in the CMB power spectrum, allow us to extract key cosmological parameters, such as the density of baryonic matter and cold dark matter components, the expansion rate of the universe, and even the nature of the initial quantum fluctuations from inflation.

The Λ Cold Dark Matter (ΛCDM) model is the standard model of cosmology. It describes the evolution and large-scale structure of the universe using six fundamental parameters. These include the baryon density (Ω_b), cold dark matter density (Ω_{cdm}), the Hubble parameter (H_0), the scalar spectral index (n_s), the amplitude of primordial fluctuations (A_s), and the optical depth due to reionization (τ_{reio}). In this project, we focus on the parameter Ω_b, which quantifies the contribution of ordinary (baryonic) matter

such as protons and neutrons to the overall energy density of the universe.

This project is divided into two parts. In the first part, we investigate how variations in each of the six ΛCDM parameters affect the Cosmic Microwave Background (CMB) temperature power spectrum. Each parameter is varied by ±10% from its Planck 2018 [1] best-fit value, and the resulting changes in the C_ℓ spectrum are analyzed. This provides an intuitive understanding of the physical role played by each parameter. In the second part, using simulated theoretical predictions from the CLASS[2] (Cosmic Linear Anisotropy Solving System) code and comparing them with Planck satellite data, we aim to find the best-fit value of Ω_b through a chi-squared analysis. To keep things simple, we vary only this one parameter while holding the others fixed, observing how changes in Ω_b affect the shape of the CMB power spectrum. This provides insight into how accurately we can determine the cosmic energy budget from just one part of the CMB data.

The chapters that follow provide a clear flow from theory to experimental analysis. Chapter 2 covers the necessary background, including key cosmic events and the physics of the CMB power spectrum. Chapter 3 discusses the methodology, including the data used, computational tools such as `CLASS` and `Healpy`, and the statistical techniques applied. Chapter 4 presents the results in two parts: first, the effects of varying each ΛCDM parameter by ±10%, and second, the determination of the baryon density Ω_b using the χ^2 minimization method. The final chapter summarizes the findings and suggests directions for future work.

Chapter 2

Theory

2.1 The ΛCDM Model

The ΛCDM model is the standard model used to describe the evolution and structure of the universe. It is based on the assumption that the universe is homogeneous and isotropic on large scales and includes contributions from ordinary matter, dark matter, and dark energy (represented by the cosmological constant Λ). The model is defined by six main parameters that can be fitted using observations like the cosmic microwave background (CMB).[3]

These parameters are: the baryon density Ω_b, cold dark matter density Ω_{cdm}, the Hubble parameter H_0, scalar spectral index n_s, amplitude of primordial fluctuations A_s, and optical depth due to reionization τ_{reio}. In this project, we particularly focus on Ω_b, and explore how its value affects the angular power spectrum of the CMB.

2.2 Timeline of Key Cosmic Events[4]

1. **Initial Stage:** The early universe was an extremely hot and dense state, consisting of a tightly coupled plasma of photons and matter. During this phase, density perturbations were seeded by quantum fluctuations in the inflationary field, a scalar field analogous to temperature variations in a room. These fluctuations generated nearly scale-invariant primordial gravitational potentials, with similar amplitudes across different wave numbers k. Pressure gradients induced by these fluctuations led to acoustic oscillations; essentially sound waves in the photon-baryon fluid. Cold dark matter (CDM), while not interacting with radiation, significantly influenced the dynamics by deepening gravitational potential wells, thereby modifying the resulting acoustic pattern.

2. **Recombination:** As the universe cooled, electrons and protons combined to form neutral hydrogen in a stable manner. This marked the decoupling of photons from

baryons, as the plasma transitioned into a neutral gas. Acoustic oscillations ceased, and the existing distribution of density perturbations became imprinted in the CMB as anisotropies. These imprints also correspond to the initial baryon acoustic oscillation (BAO) scale that is observable in the large-scale distribution of galaxies today.

3. **Dark Ages:** In the absence of radiation sources, the universe entered a period known as the Dark Ages. During this epoch, CDM began to undergo gravitational collapse in overdense regions, forming the potential wells into which baryons would later fall.

4. **Cosmic Dawn:** The formation of the first luminous sources, such as stars and early galaxies, marked the end of the Dark Ages. The radiation from these objects began to reionise the intergalactic medium, making it partially ionised once again.

5. **Structure Formation:** Over time, matter continued to collapse under gravity, forming increasingly complex structures. CDM played a major role in forming the cosmic web, an interconnected network of dark matter and galaxies. The baryonic matter, influenced by the dark matter distribution, retained the BAO correlation length originally established at recombination.

6. **Dark Energy Domination:** In the later stages of cosmic evolution, dark energy , characterized by negative pressure, began to dominate the expansion dynamics of the universe. Unlike matter, which dilutes with expansion, dark energy remains approximately constant in density. This led to the accelerated expansion of the universe, overpowering the gravitational attraction of matter.

2.3 Brief Overview of the CMB Power Spectrum [5, 6]

On large scales, the Cosmic Microwave Background (CMB) appears as a nearly perfect blackbody with a temperature of approximately 2.73 K. However, a closer look reveals small fluctuations—tiny ripples in temperature—across the sky. These anisotropies are incredibly informative, as they hold clues about the conditions of the early universe. The temperature fluctuations arise from several key physical effects occurring around the time of photon decoupling[5, 6]:

- Spatial variations in the density of the photon-baryon fluid at the Last Scattering Surface (LSS),

- Variations in the gravitational potential at the Last Scattering Surface due to the underlying matter distribution,
- Changes in gravitational potential along the CMB photon path (Integrated Sachs-Wolfe effect).

To analyze these fluctuations quantitatively, the temperature field on the sky is expanded in terms of spherical harmonics:

$$\frac{\Delta T(\hat{n})}{T} = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{n}), \tag{2.1}$$

where $\hat{n}$ represents a direction on the sky, $Y_{\ell m}$ are the spherical harmonics, and $a_{\ell m}$ are the expansion coefficients. The statistical properties of the CMB are encoded in the angular power spectrum, defined as the variance of the harmonic coefficients:

$$C_\ell = \langle |a_{\ell m}|^2 \rangle. \tag{2.2}$$

Since the CMB sky is assumed to be statistically isotropic, the expectation value depends only on the multipole moment ℓ, which corresponds to angular scale. A more commonly plotted quantity is:

$$D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell, \tag{2.3}$$

which roughly gives the power per logarithmic interval in ℓ and highlights the acoustic peaks more clearly. This formalism is adapted from Klauber's student-friendly CMB guide [6].

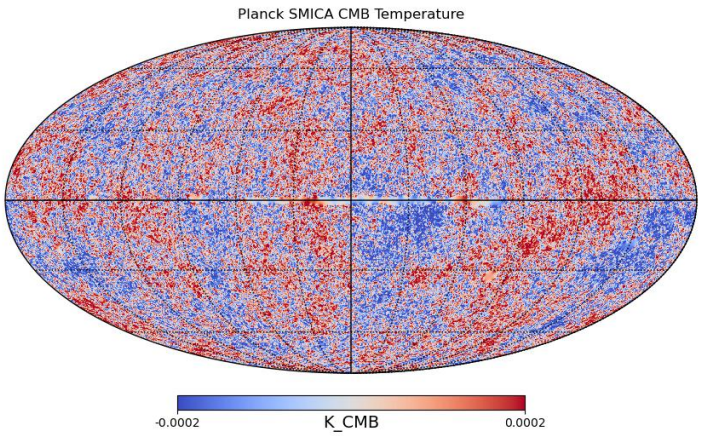

Figure 2.1: CMB temperature map produced using the `healpy` library and Planck data.

As we measure the temperature of the CMB in smaller and smaller patches of the sky, the fluctuations at different angular scales can be analyzed. The resulting CMB

power spectrum encodes physical information about the universe's contents and geometry. Importantly, the shape and location of the peaks in the power spectrum depend on the values of various cosmological parameters.

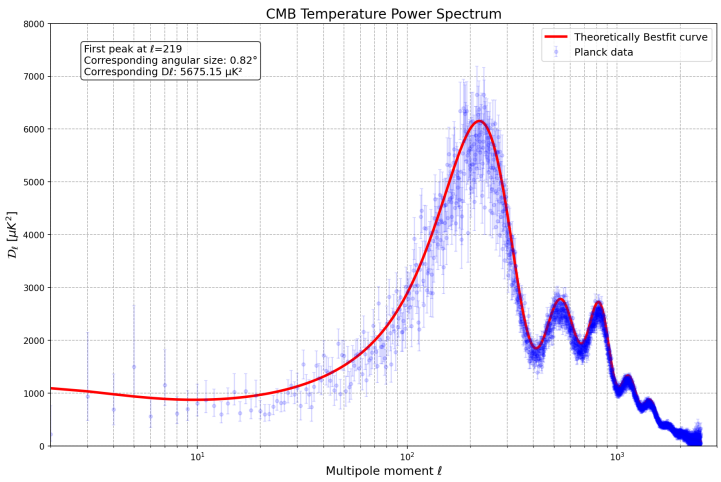

Figure 2.2: CMB temperature power spectrum using Planck data (blue) and theoretical best-fit values (red).

Key physical interpretations from the power spectrum:

- The initial flat plateau represents large-scale modes (low ℓ) that were outside the horizon at recombination and thus had not yet oscillated.
- The location of the first acoustic peak (around $\ell \sim 220$) serves as a "standard ruler" for the size of the sound horizon, helping determine spatial curvature:
  - $\ell \sim 220$: Flat universe,
  - $\ell > 220$: Open universe,
  - $\ell < 220$: Closed universe.
- Odd-numbered peaks represent compressions and are typically larger than the even-numbered (rarefaction) peaks.
- The height of the second peak is sensitive to the baryon content (Ω_b), while the third peak's height is influenced by dark matter density (Ω_{cdm}).
- A boosted third peak suggests dark matter dominated the energy density of the plasma before recombination.
- At small angular scales (large ℓ), photon diffusion (Silk damping) causes the power to drop exponentially.

Chapter 3

Methodology

3.1 Data

We used the Planck satellite's final data release for CMB temperature and polarization measurements. This included the TT, TE, and EE power spectra along with lensing reconstruction data. The multipole range covered $\ell = 2$ to 2508 for the CMB spectra.

The simulated data came from CLASS, which generated theoretical CMB spectra for different sets of cosmological parameters. In the first part, we varied each parameter by $\pm 10\%$ to understand the behaviour of the power spectrum on varying them differently. Later, to determine the value of a single parameter, we focused on varying the baryon density Ω_b while keeping other ΛCDM parameters fixed to Planck's best-fit values. The output from CLASS was processed to match the same ℓ range and units (μK^2) as the Planck data.

3.2 Codes

The analysis used two main computational tools:

CLASS was used for generating theoretical CMB spectra. We made a python code that computed lensed CMB power spectra for each set of input parameters, including temperature and polarization spectra.

Python served as our analysis platform with several key libraries:

- Healpy for reading and visualizing Planck maps
- NumPy and SciPy for numerical calculations and curve fitting
- Matplotlib for creating all figures and plots
- Astropy for handling FITS files and cosmological calculations

We used the `healpy` Python library to visualize the Cosmic Microwave Background (CMB) temperature anisotropies (refer figure 2.2). Starting with the Planck-provided `.fits` map, we used functions like `hp.read_map()` to load the data and `hp.mollview()` to plot the sky in Mollweide projection. To smooth the data and better observe the large-scale anisotropies, we applied `hp.smoothing()` and specified the beam width. This allowed us to explore the large-scale structure and gain familiarity with the CMB sky. The anisotropies were observed by transforming pixel values into temperature deviations in microkelvin.

To investigate the theoretical predictions of the ΛCDM model, we used `CLASS` (Cosmic Linear Anisotropy Solving System). CLASS is a fast and accurate Boltzmann code used to compute cosmological observables such as the CMB angular power spectrum. CLASS numerically solves the Einstein-Boltzmann equations to generate output based on a given set of cosmological parameters [2]. For each parameter, we generated spectra with deviations of ±10% from the best-fit Planck values while keeping the remaining parameters constant. This was done by editing a copy of the explanatory file with all the best-fit parameters, and running CLASS to produce the theoretical angular power spectra. The output `.dat` files were then parsed using Python, and the C_ℓ values were plotted against multipole moment ℓ to analyze how variations in each parameter affected the spectrum. This comparison helped us understand each parameter's role in shaping the anisotropy features.

A more detailed analysis was performed for the baryon density parameter Ω_b. We wrote Python scripts to generate multiple input files with varying Ω_b values and adjusted Ω_{cdm} accordingly to preserve a flat universe, while keeping all other parameters fixed to the Planck best-fit values. Bash scripting was used to run these modified files through CLASS, generating 20 theoretical spectra. Using Python, we computed the chi-squared statistic

$$\chi^2 = \sum_\ell \frac{(D_\ell^{\text{Planck}} - D_\ell^{\text{theory}})^2}{\sigma_\ell^2}$$

to compare theoretical spectra with Planck observational data. By identifying the Ω_b value corresponding to the minimum χ^2, we refined our parameter space. This process was iteratively repeated, narrowing the Ω_b range to get a more precise value. At later stages, interpolation was used to estimate the minimum more precisely. Initially, only the temperature (TT) spectra were used; eventually, we included the polarization spectra (EE, TE), which significantly improved the accuracy of our inferred Ω_b.

3.3 Statistics

To assess how well a theoretical spectrum, out of the models we simulated, fits the observational data, we employed the chi-squared (χ^2) minimization technique. For each run of CLASS with a specific value of Ω_b, the output angular power spectrum was compared to Planck data using the formula:

$$\chi^2 = \sum_{\ell=\ell_{\min}}^{\ell_{\max}} \frac{(D_\ell^{\text{Planck}} - D_\ell^{\text{theory}})^2}{\sigma_\ell^2}$$

where D_ℓ is the unbinned power spectrum (commonly expressed as $D_\ell = \ell(\ell+1)C_\ell/2\pi$), and σ_ℓ is the uncertainty in Planck's measurement at each multipole moment ℓ.

By computing χ^2 for each theoretical spectrum, we identified which Ω_b values best reproduced the observed anisotropy data. A lower χ^2 indicated a closer match. The process was repeated iteratively with finer sampling as we approached the minimum, helping to locate the best-fit baryon density. We later interpolated between the discrete points to further refine the minimum. Initially, χ^2 was calculated using only the TT spectrum, but in later stages, we incorporated EE and TE spectra as well. This multi-spectrum analysis provided stronger statistical constraints and reduced degeneracies, ultimately leading to a more precise determination of Ω_b.

Chapter 4

Results

4.1 Section A: Varying the Parameters

Matter Density Parameter Ω_m

The parameter Ω_m represents the present-day fractional energy density of matter, including both baryonic and dark matter:

$$\Omega_m = \frac{\rho_m}{\rho_{\text{crit}}}$$

Current constraints suggest $\Omega_m \approx 0.3$, indicating matter contributes roughly 30% of the total energy density of the universe. A variety of observational techniques have been used to constrain Ω_m [5, 7].

Similar to the case of n_s and τ, several methods (notably cluster abundances and weak lensing) are sensitive to a degenerate combination of Ω_m and σ_8. This degeneracy appears in the amplitude of the matter power spectrum:

$$P(k) \propto \sigma_8^2 \cdot k^{n_s - 1}$$

In such cases, increasing Ω_m can be compensated by lowering σ_8, and vice versa. To break this degeneracy, joint analysis with external data (like CMB, lensing, galaxy clustering) is required [8].

4.1.1 Baryonic Density $\Omega_b h^2$

The parameter Ω_b quantifies the present-day fractional energy density of baryonic matter (protons, neutrons). It is often expressed in terms of the physical baryon density:

$$\Omega_b h^2$$

Two observational methods are used to determine $\Omega_b h^2$ includ CMB power spectrum fits (sensitive primarily to the relative amplitudes of the first and second acoustic peaks) and Primordial deuterium abundance + BBN theory [2, 9].

Baryons interact with photons in the early universe, influencing the dynamics of acoustic oscillations (BAOs). An increase in Ω_b enhances the inertia of the photon-baryon fluid, affecting peak heights as follows:

- Odd (compression) peaks: Enhanced due to greater gravitational collapse.
- Even (rarefaction) peaks: Suppressed due to increased resistance to outward pressure.

The current known value for $\Omega_b h^2$ is 0.02242 ± 0.00014 [1]. The matter power spectrum is also affected:

$$P(k) \propto \Omega_b \cdot k^{n_s - 1}$$

On varying Ω_b as Ω_b $\pm 10\%$ (refer figure 4.1)

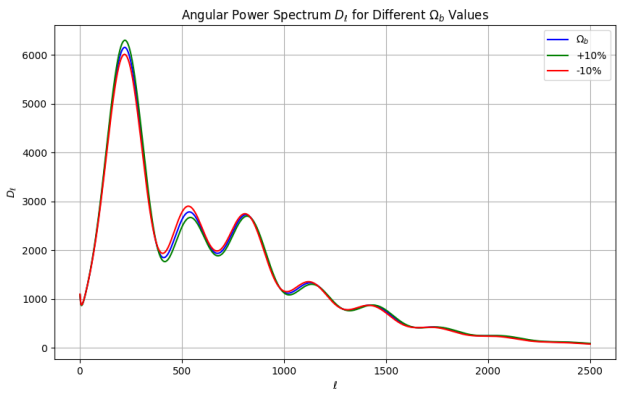

Figure 4.1: Variation in power spectra if Ω_b is varied

- $+10\%$ in Ω_b leads to increased odd peak heights and decreased even peak heights. A higher baryon density increases the inertia of the photon-baryon fluid, strengthening compressional phases and damping rarefactions.
- -10% causes the acoustic peak contrast to lessen, reducing the prominence of the first peak and raising the second. A lower baryon density leads to a more symmetric oscillation pattern and shallower gravitational potential wells, affecting the timing and structure formation process.
- We also see a slight shift in the damping tail [9]. This is because increasing the baryon density, makes the photon-baryon fluid more tightly coupled at recombination and hence shifts the damping tail to smaller angular scales or higher multipole moments l. Vice-versa for decreasing the baryon density. fig: Variation in power spectra if Ω_b is varied

4.1.2 Cold Dark Matter Density $\Omega_{\mathbf{cdm}}h^2$

The parameter Ω_{cdm} quantifies the present-day fractional energy density of non-baryonic cold dark matter (CDM). It is typically expressed in terms of the physical CDM density $\Omega_{\text{cdm}}h^2$. In ΛCDM cosmology, Ω_{cdm} is inferred primarily through fits to CMB temperature and polarization power spectra [1, 2]. CDM does not interact electromagnetically, but it leaves a gravitational imprint that affects the growth of structure and the shape of the CMB anisotropy spectrum. It contributes to the total matter budget as:

$$\Omega_{\text{cdm}} = \Omega_m - \Omega_b$$

Higher acoustic peaks are very much dependent on radiation to dark matter ratio. Since the amount of radiation is known from the measured temperature of the CMB and the thermal history, under normal assumptions the higher acoustic peaks are sensitive to the dark matter density in the universe. Cold dark matter affects the shape and amplitude of the matter power spectrum by setting the rate of structure growth and gravitational potential depth:

$$P(k) \propto \Omega_{\text{cdm}} \cdot k^{n_s - 1}$$

On varying Ω_{cdm} by ±10% (refer figure 4.2):

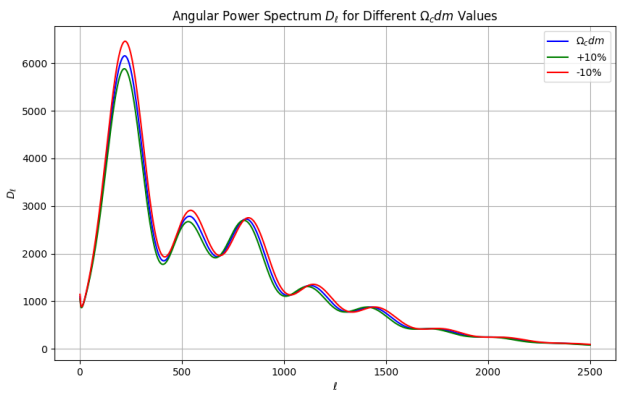

Figure 4.2: Variation in power spectra if Ω_{cdm} is varied

- +10% in Ω_{cdm} increases the gravitational potential, enhancing structure formation and boosting small-scale power [5, 7]. A higher CDM density accelerates the growth of cosmic structures by enhancing gravitational collapse, especially after recombination. We also notice that the third peak appears to be higher than the second peak; thus indicating itself as a dark matter density measure.

- −10% leads to shallower potential wells and a suppression of small-scale power in the matter power spectrum. A lower CDM density delays structure formation.

- We also note that there exists a shift in the position of the peaks, thus implying that Ω_{cdm} shall also influence the age of universe at recombination.

  Value for $\Omega_{\text{cdm}}h^2$ we know is 0.11933 ± 0.00091 [1].

4.1.3 Hubble Constant H_0

The Hubble constant, H_0, characterizes the present-day rate of expansion of the universe and sets the absolute scale of cosmic distances and timescales[4, 8]. It is the $z = 0$ value of the redshift-dependent Hubble parameter $H(z)$ defined in the FLRW metric. In the context of the ΛCDM model, it is given by:

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda}$$

where:

- Ω_m: matter density parameter,
- Ω_k: curvature density parameter,
- Ω_Λ: dark energy density parameter.

H_0 sets the age and size of the universe:

$$t_0 \approx \frac{1}{H_0}$$

for a matter-dominated universe with zero curvature and no dark energy. It controls the relationship between redshift and comoving distance, which underlies many cosmological observables such as angular diameter distances, luminosity distances, and the growth rate of structures. Current known value for H_0: 67.36 ± 0.54 km/s/Mpc [1].
Effect of variation in H_0 by $\pm 10\%$(refer figure 4.3):

- *If H_0 increases by 10%:* The universe expands faster, reducing the total time since recombination. As a result, the universe is younger and smaller at the time we observe the CMB, causing the same physical scales (sound horizon) to subtend larger angles. This shifts the peaks to smaller multipole values (lower ℓ).
- *If H_0 decreases by 10%:* The expansion rate is slower, making the universe older and larger at the time of observation. The same physical scales then subtend smaller angles on the sky, shifting the peaks to higher multipole values (larger ℓ). The longer age also allows more time for gravitational collapse to grow, slightly enhancing the even-numbered peaks.
- Larger changes in H_0 can affect the flatness of the universe as well.

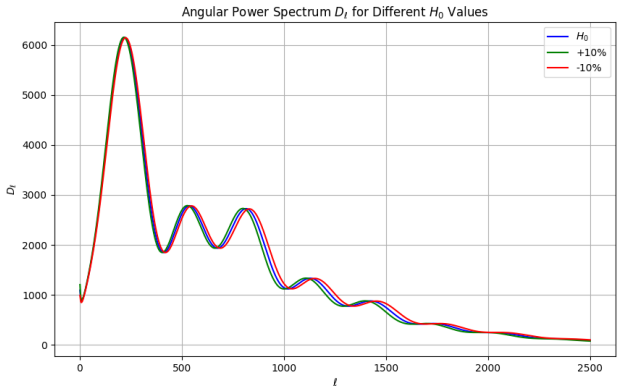

Figure 4.3: CMB power spectrum response to ±10% variation in H_0

4.1.4 Reionization Optical Depth τ

The optical depth to reionization, τ, quantifies the probability that a cosmic microwave background (CMB) photon was Thomson-scattered by free electrons released during the epoch of reionization [8, 9]. It is a unitless measure of the line-of-sight opacity to CMB photons and is defined as:

$$\tau = \int_0^{z_{\text{reion}}} n_e(z)\, \sigma_T \, \frac{dl}{dz}\, dz$$

where:

- $n_e(z)$: number density of free electrons at redshift z,
- σ_T: Thomson scattering cross-section,
- $\frac{dl}{dz}$: line element converting redshift to comoving distance.

The currently calculated value for τ is 0.0561 ± 0.0071 [1]. A larger τ implies earlier and more extended reionization (more opaque, more scattered photons), pointing to earlier star and galaxy formation. If $\tau = 0$, it would imply no reionization has occurred. Thomson scattering during reionization suppresses the temperature anisotropy of the primordial CMB on small angular scales. The amplitude of the observed power spectrum C_ℓ is damped as:

$$C_\ell^{\text{obs}} \propto e^{-\tau}\, C_\ell^{\text{primordial}}$$

This exponential damping uniformly reduces power on all scales smaller than the horizon at reionization. In contrast, reionization introduces new polarization anisotropies at large angular scales. A ±10% variation in τ has the following implications (refer figure 4.4):

- If $\tau \to 1.1\tau$, then $C_\ell^{\text{obs}} \to e^{1.1\tau} = (e^{-\tau})^{1.1}$, implying a stronger suppression. A higher τ means reionization happened earlier and the universe had more free electrons for longer, leading to increased scattering and stronger damping of the CMB anisotropies.

- If $\tau \to 0.9\tau$, then $C_\ell^{\text{obs}} \to (e^{-\tau})^{0.9}$, implying less suppression. A lower τ suggests later reionization, fewer free electrons, and weaker damping, implying that the first stars and galaxies formed later.

(can be noted: reionization suppresses the heights of the acoustic peaks uniformly while generating large angle E-mode polarization)

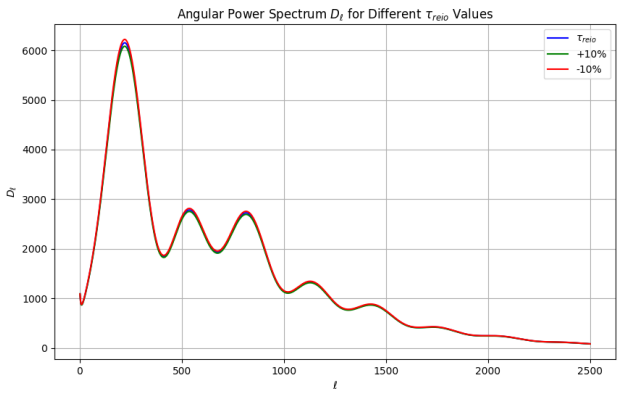

Figure 4.4: Variation in power spectra if τ is varied

4.1.5 Linear Matter Fluctuation Amplitude σ_8

The parameter σ_8 quantifies the amplitude of linear matter fluctuations on scales of $8\,h^{-1}$ Mpc at redshift $z = 0$. It measures the root-mean-square (rms) density variation in spheres of this radius and is a key descriptor of the level of matter clustering in the universe.

A higher A_s implies stronger clustering, which affects large-scale structure formation, galaxy distributions, and gravitational lensing signals[7, 9]. CMB anisotropies set the initial conditions for structure growth and the growth is governed by matter content and gravity. Multi We also note that there exists a shift in the position of the peaks, thus implying that Ω_{cdm} shall also influence the age of universe at recombination. Observational techniques are used to constrain A_s, probing different redshifts and scales. Currently known value for A_s is such that $(\ln(10^1\ A_s))$: 3.047 ± 0.014 [1].

Like Ω_m, σ_8 affects the matter power spectrum:

$$P(k) \propto \sigma_8^2 \cdot k^{n_s - 1}$$

However, the two are degenerate in certain observables (e.g., cluster counts, lensing), requiring complementary data (e.g., BAO, CMB) to break the degeneracy.
The effect of $\pm 10\%$ variation on $A_s$is as follows (refer figure 4.5):

- $+10\%$ in A_s: $P(k)$ increases, leading to stronger small-scale clustering. Higher σ_8: Faster structure formation; more massive galaxy clusters; stronger lensing signals.
- -10% in A_s: $P(k)$ decreases, reducing the clustering amplitude. Lower σ_8: Delayed formation of large-scale structures; weaker lensing and galaxy correlations.

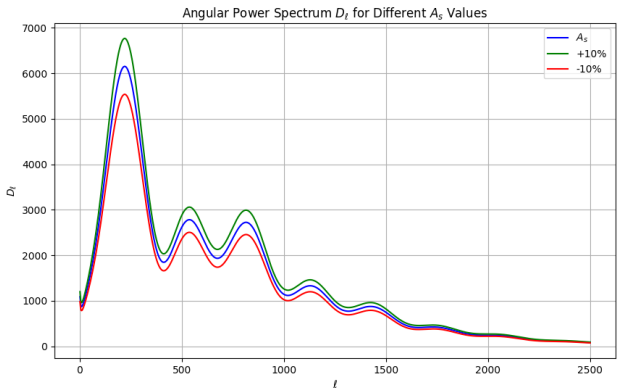

Figure 4.5: Variation in power spectra if A_s is varied

4.1.6 Scalar Power-Law Index n_s

In the ΛCDM model, the primordial scalar power spectrum is characterized by a power law, $P(k) \propto k^{n_s-1}$ [5, 9]. n_s describes how the density fluctuations vary with scale. Inflation generally predicts $n_s < 1$ due to the slowly varying potential during inflation.

Earlier CMB temperature-only measurements faced degeneracy between n_s and the optical depth to reionization, τ. This degeneracy was resolved with polarization data from WMAP and Planck as it provides a clear signal of τ. The current measurement of n_s is 0.9665 ± 0.0038, thus supporting inflationary predictions[1, 8].

Varying the scalar spectral index n_s by $\pm 10\%$ (refer figure 4.6) changes the slope of the primordial power spectrum $P(k) \propto k^{n_s-1}$.

- If n_s increases by 10%, the spectrum becomes *blue-tilted*, enhancing power on small (high-k) scales and reducing power on large scales. This implies that smaller structures (galaxies, clusters) could form earlier or in greater numbers.
- If n_s decreases by 10%, the spectrum becomes *red-tilted*, enhancing large-scale power and suppressing small-scale structure. (Small-scale fluctuations weaker $\rightarrow$ less clumpy universe early on $\rightarrow$ delayed or suppressed small-scale structure.)

This can be observed mathematically if we try to compute $^{n_s-1}$. In a nutshell, physically, this affects the formation of structures: a higher n_s implies more small-scale fluctuations, potentially leading to earlier or denser galaxy formation. A lower n_s suggests smoother early conditions, delaying structure formation on small scales.

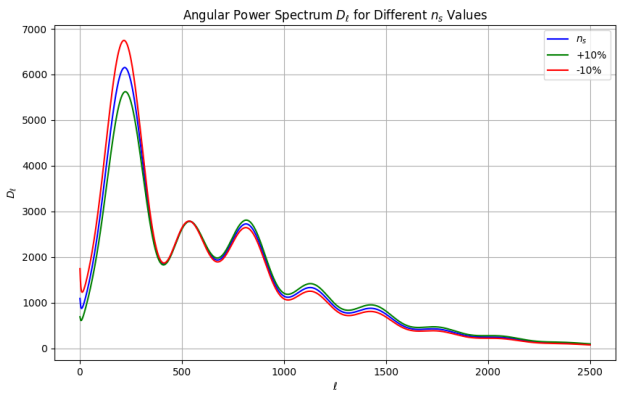

Figure 4.6: Variation in power spectra if n_s is varied

4.2 Section B: Finding Ω_b Using χ^2 Statistics

After calculating χ^2 for each Ω_b, we plotted χ^2 as a function of Ω_b. The minimum point of this curve corresponds to the best-fit value. To get a precise estimate, we fitted a parabola around the lowest χ^2 values and calculated the uncertainty as the range within which χ^2 increases by 1 from the minimum.

The plots below demonstrate various stages of zooming and data refinement to extract the best-fit Ω_b from the CMB angular power spectrum.

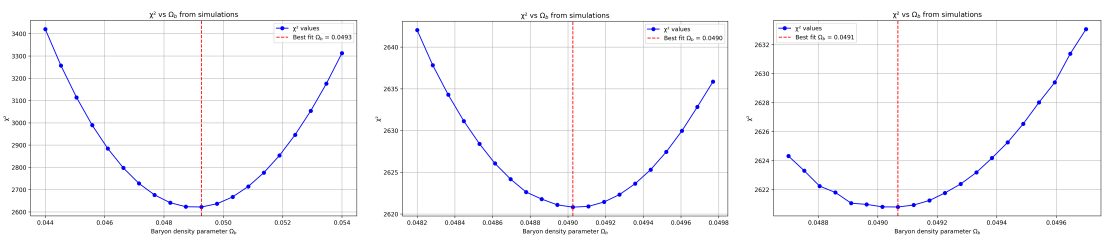

Figure 4.7: Zoomed-in views of the χ^2 vs Ω_b curve. Each successive plot narrows the Ω_b range around the minimum to get a clearer picture of the parabolic nature of the curve.

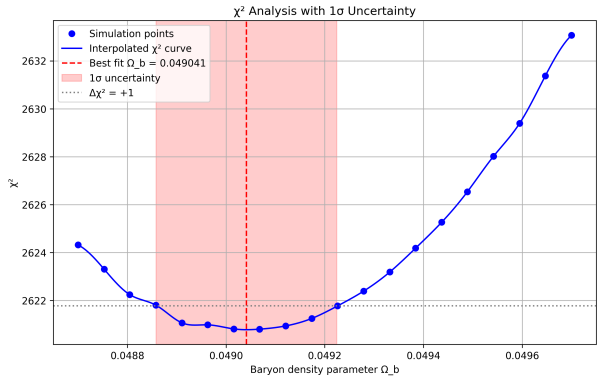

Figure 4.8: Interpolation applied to the χ^2 curve to get a smoother estimate of the minimum. This helps in identifying the best-fit Ω_b value more accurately.

Initially, we used only the TT spectrum. However, to improve accuracy and break degeneracies with other parameters like Ω_{cdm}, H_0, and n_s, we later included the TE and EE spectra as well. This gave us a more reliable and precise estimate for Ω_b.

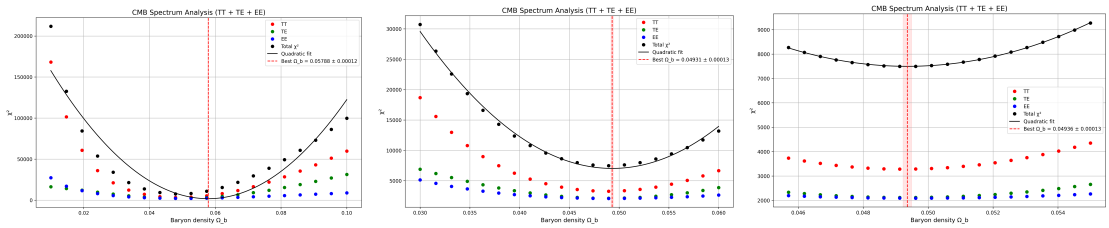

Figure 4.9: The χ^2 vs Ω_b curve using combined TT, EE, and TE datasets. Each plot shows zoomed-in views of the χ^2 minimum when all spectra are used together.

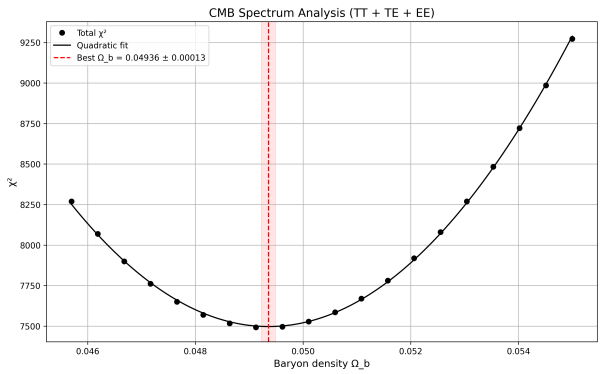

Figure 4.10: All other curves removed to isolate the clean parabolic curve from the combined dataset. Interpolation is used here again to get a precise estimate of the Ω_b that minimizes χ^2.

From the final interpolated parabolic fit, the best-fit value of Ω_b is identified with improved precision. This method ensures the curve behaves like a smooth parabola, allowing numerical methods to estimate the minimum more accurately.

The final value we obtained using this method was:

$$\Omega_b = 0.04936 \pm 0.00013$$

This result agrees very well with Planck's value when BAO data is excluded: $\Omega_b = 0.04930 \pm 0.00086$. Our smaller uncertainty is likely due to fixing the other parameters during the analysis.

Chapter 5

Summary and Future work

In this project, we studied the Cosmic Microwave Background (CMB) and how it can be used to determine the energy content of the universe. Our focus was on estimating the baryonic matter density parameter Ω_b using the angular power spectrum of CMB anisotropies.

We began by learning how to visualize and analyze the CMB using tools like `healpy`, and then used the CLASS Boltzmann code to simulate theoretical CMB spectra for different values of Ω_b. We compared these simulations with observational data from the Planck satellite.

Using Python, we computed the chi-squared statistic χ^2 to quantify how well each simulated spectrum matched the data. By finding the value of Ω_b that minimized χ^2, we were able to estimate the best-fit value of this parameter. The result we obtained was:

$$\Omega_b = 0.04936 \pm 0.00013$$

This is in excellent agreement with the Planck result without BAO: $\Omega_b = 0.04930 \pm 0.00086$.

Throughout the process, we explored how each cosmological parameter affects the shape of the CMB power spectrum. We also learned about parameter degeneracies and how including more data types (like EE and TE spectra) can help break these degeneracies and improve the accuracy of our estimates.

While our analysis produced a good estimate for Ω_b, there are several improvements that can be made in future studies:

- **Use of MCMC Sampling:** In our work, we varied only one parameter at a time. A more realistic approach would be to use Markov Chain Monte Carlo (MCMC)

methods to explore the full parameter space. This would allow for joint estimation of Ω_b, Ω_{cdm}, H_0, and n_s, taking into account their correlations.

- **Inclusion of More Data:** Our current analysis used only the CMB power spectra. Including other datasets such as BAO would provide stronger constraints.

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